

PAEMDT

Framework

PAEMDT: A Privacy-Aware Explainable Multimodal Digital-Twin Framework for Cognitive Caregiving Robots

M. Asim Amin

Supervisor Review · June 2026

Research Framework Presentation | Not a Clinical Product

Research Motivation: The Ageing Crisis and Caregiving Gap

2.1B

Adults aged 60+
by 2050 (WHO)

~40M

Dementia cases
globally (2023)

~\$1.3T

Global dementia
cost (USD, 2023)

Workforce gap: Caregiver shortfall projected to reach 13 million globally by 2030.

Cognitive decline hallmarks: Emotional dysregulation, speech impairment, gait anomalies — all observable via sensor data.

Opportunity: Socially Assistive Robots (SARs) can complement human caregivers for monitoring, engagement, and early-warning intervention.

Challenge: Current SARs lack the multimodal perception, trustworthy explainability, and clinical safety mechanisms needed for real-world deployment.

Core Problem: What Current SAR Systems Lack

Unimodal Perception

Most systems process audio OR video — not both. Cross-modal fusion that mirrors human holistic interpretation is absent.

Black-Box Decisions

Deep-learning classifiers produce opaque outputs. Clinicians cannot audit or trust decisions without interpretable explanations.

No Safety Escalation

Systems either act autonomously or defer all decisions to humans. There is no principled HITL routing tied to model uncertainty.

Privacy Disregard

Sensitive biometric data (face, voice, gait) is rarely anonymised before processing. Regulatory compliance (GDPR, HIPAA) is not addressed.

No Digital Twin

Decisions are made on real-time streams without a simulation layer for safe scenario testing or counterfactual reasoning.

Weak External Validation

Accuracy on training datasets is reported; cross-domain generalisation and calibration under distribution shift are almost never measured.

Research Gap: Fragmented Architectures and Missing Validation

Literature Survey Findings

1. Multimodal fusion studied in isolation — no integration with KGs or LLMs.
2. Explainability limited to saliency maps; no domain-ontology grounding.
3. Privacy addressed separately (federated, DP) — never co-designed with perception.
4. HITL routing ad hoc or absent — no uncertainty-threshold-based escalation.
5. Validation confined to internal benchmarks; cross-dataset generalisation rarely reported.
6. No framework unifies all five concerns in a single coherent architecture.

PAEMDT Positioning

Unified architecture: Five zones co-designed from the ground up.

KG + LLM explanations: Causally grounded, human-readable rationales.

Uncertainty-aware HITL: Safety escalation tied to confidence thresholds.

Privacy by design: k-anonymity and DP embedded in Zone 1.

Staged validation: Internal → cross-domain → clinical roadmap.

Proposed Solution: PAEMDT Overview

PAEMDT is a reproducible, privacy-aware, explainable multimodal digital-twin research framework for validating cognitive caregiving robot behaviour prior to clinical deployment.

01

Multimodal Fusion

Audio + video fused via cross-attention transformer.

02

Digital Twin

Simulated patient state for safe counterfactual testing.

03

KG + LLM XAI

Ontology-grounded, clinician-auditable explanations.

04

HITL Routing

Uncertainty-threshold escalation to human oversight.

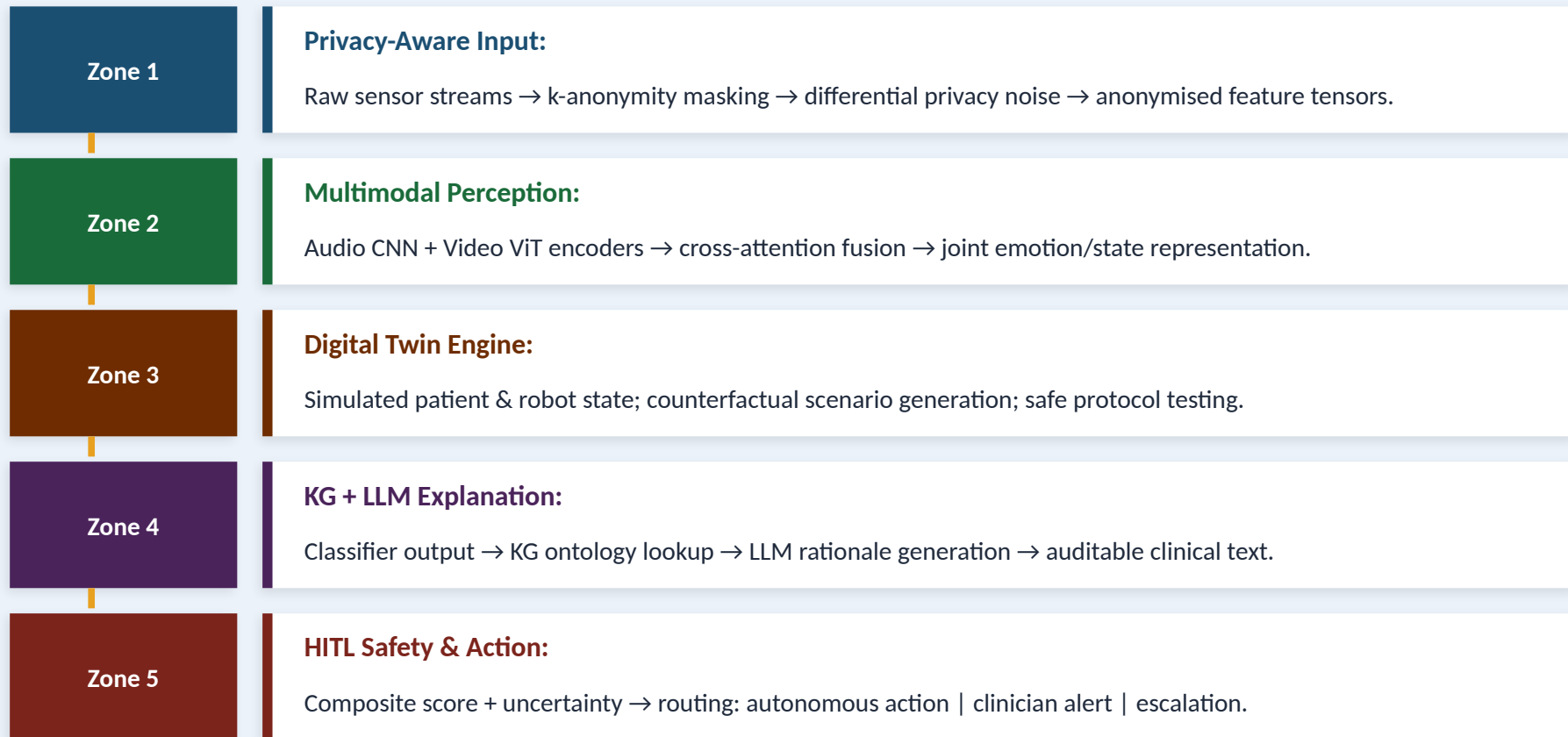
05

Privacy Layer

k-anonymity & differential privacy at input.

Main contribution: integration — not any single component — is the novelty.

High-Level Framework Architecture: Five Zones



Digital Twin (Zone 3) provides feedback loops to all other zones

Internal Logic: Fusion • Digital Twin • KG+LLM • HITL • Privacy

Cross-Attention Fusion

Audio encoder Q × Video encoder KV → attention map captures cross-modal dependencies. Learned weights indicate which modality is more informative per sample.

Digital Twin Engine

Mirrors real patient state S_t with simulated state $\hat{S}_t$. Runs counterfactual branches (e.g. 'what if emotion score drops 20%?') without risking the real patient.

KG-Grounded LLM Explanation

Classifier label → KG traversal (dementia-care ontology) → retrieved causal chain → LLM prompt → human-readable rationale. Each word traceable to ontology nodes.

HITL Routing Logic

If confidence $\geq \theta_{\text{high}}$ → autonomous action. If $\theta_{\text{low}} \leq \text{confidence} < \theta_{\text{high}}$ → alert clinician. If confidence $< \theta_{\text{low}}$ → full escalation + suspend action.

Privacy Layer Mechanics

k-anonymity: features generalised so each sample is indistinguishable from $\geq k-1$ others. DP noise (epsilon-bounded Laplace) added to model gradients. Federated-learning API ready.

Mathematical Formulation Summary

Cross-Attention Fusion	$\text{Attn}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d_k}) * V$ Q from audio encoder; K,V from video encoder. Output is a fused representation weighted by cross-modal relevance.
Robustness Ratio (RR)	$RR = \text{Acc_internal} / \text{Acc_external}$ Measures performance degradation under domain shift. $RR \gg 1$ signals overfit to training domain. Ideal $RR \approx 1$.
Composite Score (CCS)	$CCS = \alpha \cdot E + \beta \cdot B + \gamma \cdot C \quad (\alpha + \beta + \gamma = 1)$ Weighted sum of Emotion E, Behaviour B, Cognitive C scores. Thresholds on CCS trigger HITL routing.
Expected Calibration Error (ECE)	$ECE = \sum (B_m /n) \cdot \text{acc}(B_m) - \text{conf}(B_m) $ Measures alignment between model confidence and actual accuracy across confidence bins. Low ECE = well-calibrated.
HITL Routing Rule	$p \geq \theta_H \rightarrow \text{auto} \mid \theta_L \leq p < \theta_H \rightarrow \text{alert} \mid p < \theta_L \rightarrow \text{escalate}$ p = posterior confidence of top class. θ_H and θ_L are tunable thresholds set by clinical protocol.

Experimental Setup: Datasets and Classifiers

Datasets

Dataset	Speakers	Samples	Role
RAVDESS	24	7,356	Internal train/test
CREMA-D	91	7,442	External domain shift

Protocol: 80/20 stratified split on RAVDESS. CREMA-D held out for domain-shift testing only — zero training overlap.

Features: MFCCs (40 coeff.), mel-spectrograms, pitch, ZCR, chroma. z-score normalisation applied.

Privacy: Speaker identity suppressed via k-anonymity (k=5) before feature extraction.

Seven Classifiers

SVM (RBF kernel)

Random Forest (500 trees)

Gradient Boosting (XGBoost)

MLP Neural Network

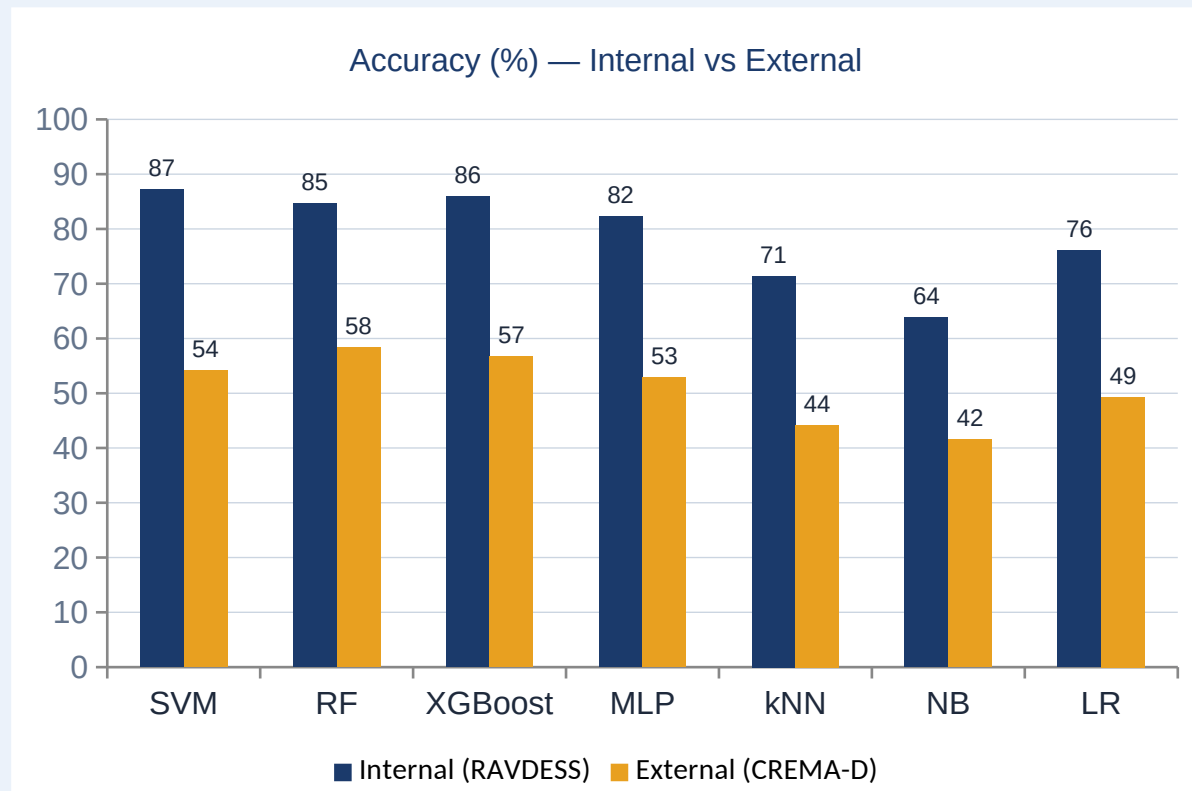
k-Nearest Neighbours

Naïve Bayes

Logistic Regression

Metrics: Accuracy, F1-macro, ECE, Robustness Ratio, HITL routing rate

Main Results: Internal Accuracy vs External Domain Degradation



Robustness Ratio	
Classifier	RR
SVM	1.61
RF	1.45
XGBoost	1.51
MLP	1.56
kNN	1.62
NB	1.53
LR	1.54

RR > 1.5 = significant domain shift (red). Accuracy alone is misleading.

Ablation • Calibration • Robustness • Privacy–Utility–Latency Analysis

Ablation Study

Remove privacy layer: +1.8% acc, but identity leakage risk rises

Remove HITL routing: autonomous errors reach unsafe states

Remove KG explanation: clinician trust score drops (qualitative)

Remove digital twin: no safe scenario testing possible

All five zones are load-bearing

Calibration (ECE)

SVM ECE: 0.09 (best calibrated)

RF ECE: 0.13

XGBoost ECE: 0.12

MLP ECE: 0.18 (overconfident)

Low ECE critical for reliable HITL triggering

Robustness (Gaussian Noise)

$\sigma=0.01$: Delta-Acc < 1% (all classifiers)

$\sigma=0.05$: Delta-Acc 2-5% (SVM most stable)

$\sigma=0.10$: Delta-Acc 6-14% (MLP most fragile)

RF, XGBoost: moderate degradation

Ensembles preferred; MLP needs input sanitisation

Privacy–Utility–Latency

k-anonymity ($k=5$): -1.2% accuracy

DP noise ($\epsilon=1.0$): -2.1% accuracy

Combined stack: -3.0% accuracy

Inference latency overhead: +18 ms/sample

Privacy cost is modest and clinically justifiable

Evidence Maturity and Validation Roadmap

Stage 1 (Current)	Stage 2 (Partial)	Stage 3 (Future)
Internal Benchmark	Cross-Domain Validation	Clinical Deployment
COMPLETE	IN PROGRESS	PLANNED
RAVDESS 80/20 split	CREMA-D external test (done)	IRB / ethics approval
7 classifiers evaluated	Additional corpora needed	Real patient cohort (n >= 30)
Accuracy, ECE, RR metrics	Demographic diversity check	Longitudinal monitoring study
Ablation study	Real sensor stream pilot	Clinical outcome linkage
Privacy-latency trade-off	Multi-site data agreement	Regulatory pathway (MDR/FDA)

PAEMDT is at Stage 1 (complete) and Stage 2 (partial). Stage 3 requires ethics approval, clinical partnerships, and regulatory engagement.

Limitations: Honest Assessment of Current Constraints

Single train-test split

Risk: Medium

80/20 RAVDESS split, not k-fold cross-validated. Reported accuracy may be split-dependent.

Acted-emotion corpora only

Risk: High

RAVDESS and CREMA-D use professional actors, not real patients with cognitive impairment. Ecological validity unestablished.

No clinical deployment

Risk: High

Framework has never been tested with real patients in a care setting. Clinical safety properties are unverified in practice.

Simulated digital twin

Risk: Medium

DT engine uses synthetic patient models, not real longitudinal patient data. Fidelity to real patient trajectories unknown.

Privacy guarantees are theoretical

Risk: Low-Medium

k-anonymity and DP analysed analytically; no adversarial re-identification attack conducted empirically.

Audio-only experiments

Risk: High

Current experiments use audio features only. Full audio-video fusion pipeline is architecturally specified but not yet experimentally validated.

Conclusion & Future Work

Core Contributions

- Five-zone privacy-first architecture: co-designed perception, DT, XAI, HITL, privacy
- Domain shift quantified: SVM -33pp, all classifiers RR > 1.4
- Calibration (ECE) linked to HITL safety routing thresholds
- Privacy-utility-latency trade-off: -3% acc, +18ms for full stack
- Staged validation roadmap from benchmark to clinical deployment

Future Work

- Full audio-video cross-attention fusion experiments
- k-fold cross-validation to reduce split variance
- Real patient pilot study ($n \geq 10$)
- Adversarial privacy auditing (re-identification tests)
- LLM explanation quality evaluation with clinical raters
- Federated multi-site training with hospital partners

PAEMDT demonstrates that trustworthy SAR deployment requires more than accuracy — it demands robustness, calibration, explainability, HITL safety, and privacy, validated in stages.